Regression Model

A Simple Question

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  Ans: 51

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- 1, 3, 9, 19, 33, ... What is the next number?
  Ans: 51 $2x^2 + 1$

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- Find a pattern from the examples
  - (function $f(x) = 2x^2 + 1$ or model the data)

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How do we design a computational procedure?

A Simplified View of ML

Data
(1,3,5,7,9 ..) → **Algorithms** → $f(x)$

$$
\begin{aligned}
&Eg:\\
&int \quad function(int\, x[\;])\{\\
&\ldots\\
&\}
\end{aligned}
$$

A Simple Question (Cont.)

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- We know: 1, 3, 9, 19, 33, ... What is the next number?

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  Ans: 51 $2x^2 + 1$

A Simple Question (Cont.)

- We know: 1, 3, 9, 19, 33, ... What is the next number?
  Ans: 51 $2x^2 + 1$
- 0.99, 3.02, 9.00, 18.98, 33.01, ... What next?

A Simple Question (Cont.)

Consider a series of 2D points
(1,3), (2,6), (3,9), (4,12), ...
What is the next point?

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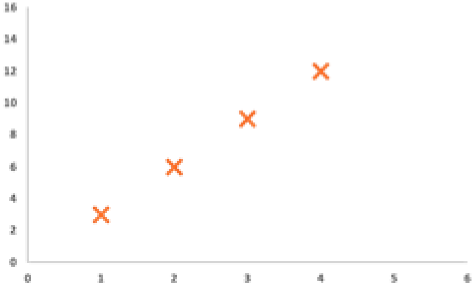
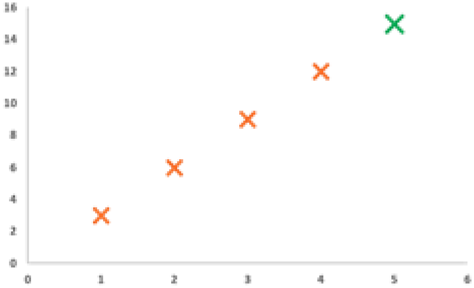


A Simple Question (Cont.)

Consider a series of 2D points
(1,3), (2,6), (3,9), (4,12), ...
What is the next point?

$(x, 3x)$ or
The function: $y = f(x) = 3x$



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When numbers are "uncertain"

- Noise in measurements
- Missing values

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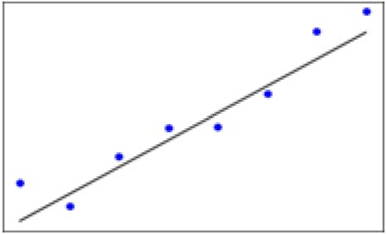
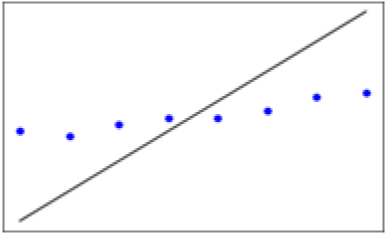
When numbers are not just "simple numbers"

- 2D points, 3D points 100
- Dimensional points

Which line is Better (Fitting)?





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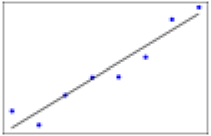

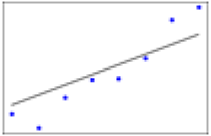

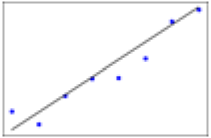

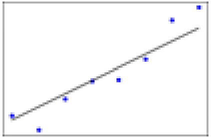

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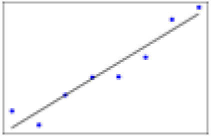
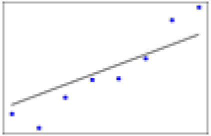
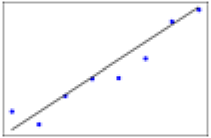


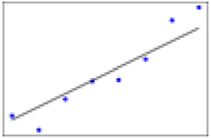
W= (1.92, 1.71)



W= (1.15, 2.71)



W= (2.12, 2.91)



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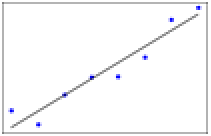
Which line is Better (Fitting)?



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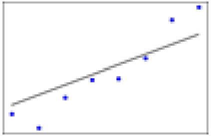

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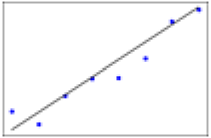
How do we Decide?



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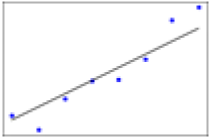

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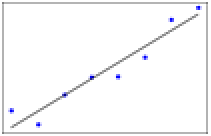
Arrange in Descending Order of Error



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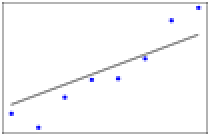
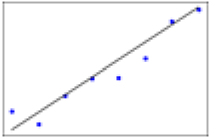
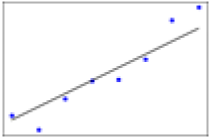


W= (1.15, 2.71)

Can we further decrease the error metric chosen?



W= (2.12, 2.91)



W= (1.48, 1.61)

Looking Further

- Principle is to decrease the loss, say Mean Square error
- Iterative algorithms like Gradient descent for that purpose

Experiments

- Demo_Linear_Regression_Pendulumn
  - Dataset : Handmade dataset (where $L \propto T^2$)
  - Objective : To apply linear regression on the dataset
- Demo_Linearregression_Diabetes
  - Dataset : Diabetes
  - Objective : To find the best fit line

Thank you!

Questions?